

Amath 483 – Homework 8

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Problem 1

Solve using, $\mathcal{F}[f(x)](\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$.

(a)

When $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \rightarrow \mathcal{F}[f](\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{1}{\sigma\sqrt{2\pi}}\right) \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] e^{-i\omega x} dx$.

Pulling out the constant factor, $\mathcal{F}[f](\omega) = \frac{1}{\sigma\sqrt{2\pi}\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \exp[-i\omega x] dx$

First combine the exponents, $\exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \exp[-i\omega x]$

$$-\frac{(x-\mu)^2}{2\sigma^2} - i\omega x = -\frac{1}{2\sigma^2} \left[(x-\mu)^2 + 2i\omega\sigma^2(x-\mu) + (i\omega\sigma^2)^2 \right] - i\omega\mu - \frac{\sigma^2\omega^2}{2} = -\frac{1}{2\sigma^2} (x-\mu+i\omega\sigma^2)^2 - i\omega\mu - \frac{\sigma^2\omega^2}{2}$$

$$\rightarrow \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \exp[-i\omega x] = \exp[-i\omega\mu] \exp\left[-\frac{1}{2\sigma^2} (x-\mu+i\omega\sigma^2)^2\right] \exp\left[-\frac{\sigma^2\omega^2}{2}\right]$$

$$\int_{-\infty}^{\infty} \exp[-i\omega\mu] \exp\left[-\frac{1}{2\sigma^2} (x-\mu+i\omega\sigma^2)^2\right] \exp\left[-\frac{\sigma^2\omega^2}{2}\right] dx = e^{-i\omega\mu} e^{-\frac{\sigma^2\omega^2}{2}} \int_{-\infty}^{\infty} \exp\left[-\frac{(x-\mu+i\omega\sigma^2)^2}{2\sigma^2}\right] dx$$

change the variables $\rightarrow u = x - \mu + i\omega\sigma^2, \quad dx = du$

When $a = \frac{1}{2\sigma^2}, \quad a \neq \pm i\omega$

$$[e^{-i\omega\mu}] [e^{-\frac{\sigma^2\omega^2}{2}}] \int_{-\infty}^{\infty} \exp\left[-\frac{(x-\mu+i\omega\sigma^2)^2}{2\sigma^2}\right] dx = [e^{-i\omega\mu}] [e^{-\frac{\sigma^2\omega^2}{2}}] \int_{-\infty}^{\infty} e^{-\frac{u^2}{2\sigma^2}} du = [e^{-i\omega\mu}] [e^{-\frac{\sigma^2\omega^2}{2}}] \sqrt{2\pi} \sigma$$

$$\mathcal{F}[f](\omega) = \frac{1}{\sigma\sqrt{2\pi}\sqrt{2\pi}} [e^{-i\omega\mu}] [e^{-\frac{\sigma^2\omega^2}{2}}] (\sqrt{2\pi} \sigma) = \frac{1}{\sqrt{2\pi}} [e^{-i\omega\mu}] [e^{-\frac{\sigma^2\omega^2}{2}}]$$

Final Answer: $\mathcal{F}[f(x)](\omega) = \frac{1}{\sqrt{2\pi}} [e^{-i\omega\mu}] [e^{-\frac{\sigma^2\omega^2}{2}}] \quad \text{when } a = \frac{1}{2\sigma^2}, \quad a \neq \pm i\omega$

(b)

When $f(t) = \sin(\omega_0 t) \rightarrow \mathcal{F}[f](\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sin(\omega_0 t) e^{-i\omega t} dt$

First, express $\sin(\omega_0 t)$ with complex exponential: $\sin(\omega_0 t) = \frac{e^{i\omega_0 t} - e^{-i\omega_0 t}}{2i}$

Substituting into the transform definition:

$$\mathcal{F}[\sin(\omega_0 t)](\omega) = \frac{1}{2i\sqrt{2\pi}} \left[\int_{-\infty}^{\infty} e^{i\omega_0 t} e^{-i\omega t} dt - \int_{-\infty}^{\infty} e^{-i\omega_0 t} e^{-i\omega t} dt \right]$$

Use the identity $\int_{-\infty}^{\infty} e^{-i\alpha t} dt = 2\pi \delta(\alpha)$,

$$\int_{-\infty}^{\infty} e^{-i(\omega-\omega_0)t} dt = 2\pi \delta(\omega-\omega_0) \text{ and } \int_{-\infty}^{\infty} e^{-i(\omega+\omega_0)t} dt = 2\pi \delta(\omega+\omega_0)$$

$$\longrightarrow \mathcal{F}[\sin(\omega_0 t)](\omega) = \frac{1}{2i\sqrt{2\pi}} \left[2\pi \delta(\omega - \omega_0) - 2\pi \delta(\omega + \omega_0) \right]$$

$$\mathcal{F}[\sin(\omega_0 t)](\omega) = \frac{2\pi}{2i\sqrt{2\pi}} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$$

$$\text{As } \frac{2\pi}{2i\sqrt{2\pi}} = \frac{\sqrt{2\pi}}{2i},$$

Final Answer: $\mathcal{F}[f(t)](\omega) = \frac{\sqrt{2\pi}}{2i} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] \quad \text{when } a = \frac{1}{2\sigma^2}, \quad a \neq \pm i\omega$

(c)

$$\text{When } f(x) = e^{-a|x|} \text{ with } a > 0 \longrightarrow \mathcal{F}[f](\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-a|x|} e^{-i\omega x} dx$$

$$\int_{-\infty}^{\infty} e^{-a|x|} e^{-i\omega x} dx = \int_{-\infty}^0 e^{-a(-x)} e^{-i\omega x} dx + \int_0^{\infty} e^{-ax} e^{-i\omega x} dx$$

$$\text{For } x < 0, \quad \int_{-\infty}^0 e^{-a(-x)} e^{-i\omega x} dx = \int_{-\infty}^0 e^{ax} e^{-i\omega x} dx = \int_{-\infty}^0 e^{x(a-i\omega)} dx$$

$$\text{For } x > 0, \quad \int_0^{\infty} e^{-ax} e^{-i\omega x} dx = \int_0^{\infty} e^{-x(a+i\omega)} dx$$

$$\text{This gives, } \int_{-\infty}^{\infty} e^{-a|x|} e^{-i\omega x} dx = \int_{-\infty}^0 e^{x(a-i\omega)} dx + \int_0^{\infty} e^{-x(a+i\omega)} dx$$

$$\text{Since } \Re(a - i\omega) = a > 0, \quad \int_{-\infty}^0 e^{x(a-i\omega)} dx = \left[\frac{e^{(a-i\omega)x}}{a-i\omega} \right]_{x=-\infty}^0 = \frac{1}{a-i\omega}$$

$$\text{Since } \Re(a + i\omega) = a > 0, \quad \int_0^{\infty} e^{-(a+i\omega)x} dx = \left[-\frac{e^{-(a+i\omega)x}}{a+i\omega} \right]_{x=0}^{\infty} = \frac{1}{a+i\omega}$$

This tells $a \neq \pm i\omega$ because the denominator should not be 0.

$$\text{Therefore, } \int_{-\infty}^{\infty} e^{-a|x|} e^{-i\omega x} dx = \frac{1}{a-i\omega} + \frac{1}{a+i\omega} = \frac{(a+i\omega) + (a-i\omega)}{(a-i\omega)(a+i\omega)} = \frac{2a}{a^2 + \omega^2}$$

$$\text{In conclusion, } \mathcal{F}[f(x)](\omega) = \frac{1}{\sqrt{2\pi}} \cdot \frac{2a}{a^2 + \omega^2} = \frac{2a}{\sqrt{2\pi}(a^2 + \omega^2)}$$

Final Answer: $\mathcal{F}[f(x)](\omega) = \frac{2a}{\sqrt{2\pi}(a^2 + \omega^2)} \quad \text{when } a \neq \pm i\omega$

(d)

$$\text{When } f(t) = \delta(t), \quad \mathcal{F}[f](\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \delta(t) e^{-i\omega t} dt$$

$$\text{Use the sifting property of the Dirac delta: } \int_{-\infty}^{\infty} \delta(t) e^{-i\omega t} dt = e^{-i\omega \cdot 0} = 1$$

$$\text{Therefore, } \mathcal{F}[\delta(t)](\omega) = \frac{1}{\sqrt{2\pi}} \times 1 = \frac{1}{\sqrt{2\pi}}$$

Final Answer: $\mathcal{F}[f(t)](\omega) = \frac{1}{\sqrt{2\pi}}$

Problem 2

$$\text{When } p(t) = \begin{cases} 0 & t < 0 \\ 1 & 0 < t < 1 \\ 0 & t > 1 \end{cases} \quad q(t) = \begin{cases} 0 & t < 0 \\ 1-t & 0 < t < 1 \\ 0 & t > 1 \end{cases}$$

$$\text{we compute } p \odot q = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} p^*(\tau) q(t + \tau) d\tau$$

Because $p(\tau) \neq 0$ only for $0 < \tau < 1$ and $q(t+\tau) \neq 0$ only when $0 < t+\tau < 1$, the integrand $p(\tau)q(t+\tau) = 1 \cdot [1 - (t+\tau)] = 1 - t - \tau$ is nonzero only for those τ satisfying simultaneously $0 < \tau < 1$ and $0 < t+\tau < 1 \iff -t < \tau < 1-t$ and the effective limits of integration are $\tau \in [\max(0, -t), \min(1, 1-t)]$.

1. When $-1 < t < 0$, then $-t > 0$ and $1-t > 1$

$$\longrightarrow \max(0, -t) = -t \quad \min(1, 1-t) = 1$$

$$\text{Therefore } p \odot q = \frac{1}{\sqrt{2\pi}} \int_{\tau=-t}^1 [1-t-\tau] d\tau$$

$$\text{Compute the integral: } \int_{-t}^1 (1-t-\tau) d\tau = \left[(1-t)\tau - \frac{\tau^2}{2} \right]_{\tau=-t}^1$$

$$\text{At } \tau = 1, (1-t) \cdot 1 - \frac{1^2}{2} = 1-t-\frac{1}{2} = \frac{1}{2}-t$$

$$\text{At } \tau = -t, (1-t)(-t) - \frac{(-t)^2}{2} = -(1-t)t - \frac{t^2}{2} = -t + t^2 - \frac{t^2}{2} = -t + \frac{t^2}{2}$$

$$\longrightarrow \int_{-t}^1 (1-t-\tau) d\tau = \left(\frac{1}{2}-t \right) - \left(-t + \frac{t^2}{2} \right) = \frac{1}{2}-t+t-\frac{t^2}{2} = \frac{1-t^2}{2}$$

$$\text{Thus, for } -1 < t < 0 \implies p \odot q = \frac{1}{\sqrt{2\pi}} \cdot \frac{1-t^2}{2} = \frac{1-t^2}{2\sqrt{2\pi}}$$

2. When $0 < t < 1$, then $-t < 0$ and $1-t < 1$

$$\longrightarrow \max(0, -t) = 0 \quad \min(1, 1-t) = 1-t$$

$$\text{Therefore } p \odot q = \frac{1}{\sqrt{2\pi}} \int_0^{1-t} [1-t-\tau] d\tau$$

$$\text{Compute the integral: } \int_0^{1-t} (1-t-\tau) d\tau = \left[(1-t)\tau - \frac{\tau^2}{2} \right]_0^{1-t} = (1-t)(1-t) - \frac{(1-t)^2}{2} = \frac{(1-t)^2}{2}$$

$$\text{In conclusion, for } 0 \leq t < 1 \implies p \odot q = \frac{1}{\sqrt{2\pi}} \cdot \frac{(1-t)^2}{2} = \frac{(1-t)^2}{2\sqrt{2\pi}}$$

3. When $t < -1$ or $t > 1$, the intervals $[0, 1]$ and $[-t, 1-t]$ do not overlap, so $p \odot q = 0$.

Final Answer:
$$p \odot q = \frac{1}{\sqrt{2\pi}} \begin{cases} \frac{1-t^2}{2} & -1 < t < 0 \\ \frac{(1-t)^2}{2} & 0 < t < 1 \\ 0 & \text{otherwise} \end{cases}$$

Problem 3

$$\text{When } p(t) = \begin{cases} 0 & t < 0 \\ 1 & 0 < t < 1, \\ 0 & t > 1 \end{cases}, \quad p \odot p = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} p^*(\tau) p(t+\tau) d\tau$$

Since $p(\tau) = 1$ exactly for $0 < \tau < 1$, and $p(t+\tau) = 1$ exactly when $0 < t+\tau < 1$, $p(\tau)p(t+\tau) = 1 \times 1 = 1$ only holds for τ satisfying both $0 < \tau < 1$ and $0 < t+\tau < 1 \iff -t < \tau < 1-t$.

Hence the integral is nonzero precisely on the intersection of $(0, 1)$ with $(-t, 1-t)$. In other words, the effective limits of integration are $\tau \in [\max(0, -t), \min(1, 1-t)]$, provided $\max(0, -t) < \min(1, 1-t)$. Outside that range, the integrand vanishes and the autocorrelation is zero.

1. When $t < -1$, then $-t > 1$ and $1 - t > 2$

$$\rightarrow \max(0, -t) = -t > 1, \quad \min(1, 1 - t) = 1$$

Since $-t > 1$, we have $\max(0, -t) > \min(1, 1 - t)$, so there is no overlap. Hence $p \odot p = 0$ for $t < -1$.

2. When $-1 < t < 0$, then $0 < -t < 1$ and $1 - t > 1$

$$\rightarrow \max(0, -t) = -t, \quad \min(1, 1 - t) = 1.$$

Therefore the integration limits become $\tau \in [-t, 1]$, and

$$p \odot p = \frac{1}{\sqrt{2\pi}} \int_{\tau=-t}^1 1 \, d\tau = \frac{1}{\sqrt{2\pi}} [\tau]_{\tau=-t}^1 = \frac{1}{\sqrt{2\pi}} (1 - (-t)) = \frac{1+t}{\sqrt{2\pi}}$$

Thus for $-1 < t < 0$, $p \odot p = \frac{1+t}{\sqrt{2\pi}}$.

3. When $0 < t < 1$, then $-t < 0$ and $0 < 1 - t < 1$

$$\rightarrow \max(0, -t) = 0, \quad \min(1, 1 - t) = 1 - t.$$

Hence the integration limits become $\tau \in [0, 1 - t]$, and

$$p \odot p = \frac{1}{\sqrt{2\pi}} \int_{\tau=0}^{1-t} 1 \, d\tau = \frac{1}{\sqrt{2\pi}} [\tau]_0^{1-t} = \frac{1-t}{\sqrt{2\pi}}$$

In conclusion, for $0 \leq t < 1$, $(p \odot p)(t) = \frac{1-t}{\sqrt{2\pi}}$

4. When $t > 1$, then $-t < -1 < 0$ and $1 - t < 0$

$\rightarrow \max(0, -t) = 0, \quad \min(1, 1 - t) = 1 - t < 0$, and therefore $\max(0, -t) > \min(1, 1 - t)$, implying no overlap. Hence $p \odot p = 0$ for $t > 1$.

Final Answer:
$$p \odot p = \frac{1}{\sqrt{2\pi}} \begin{cases} 1+t & -1 < t < 0 \\ 1-t & 0 < t < 1 \\ 0 & \text{otherwise} \end{cases}$$

Problem 4

(a)

Start with the diffusion equation: $\frac{\partial T}{\partial t}(x, t) = \kappa \frac{\partial^2 T}{\partial x^2}(x, t) \quad -\infty < x < \infty \quad t > 0 \quad T(x, 0) = T_0(x)$

1. Define the Fourier transform of $T(x, t)$

$$\tau(k, t) = \mathcal{F}\{T(\cdot, t)\}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} T(x, t) e^{-ikx} \, dx, \quad -\infty < k < \infty$$

2. Differentiate under the integral to transform $\partial_t T$

$$\frac{\partial}{\partial t} \tau(k, t) = \frac{\partial}{\partial t} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} T(x, t) e^{-ikx} \, dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{\partial T}{\partial t}(x, t) e^{-ikx} \, dx$$

Since $\partial_t T = \kappa \partial_{xx} T$, substitute:

$$\frac{\partial \tau}{\partial t}(k, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\kappa \frac{\partial^2 T}{\partial x^2}(x, t) \right) e^{-ikx} dx = \kappa \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{\partial^2 T}{\partial x^2}(x, t) e^{-ikx} dx.$$

3. Use the Fourier-transform identity for $\partial_{xx} T$

$$\int_{-\infty}^{\infty} \frac{\partial^2 T}{\partial x^2}(x, t) e^{-ikx} dx = (-ik)^2 \tau(k, t) = -k^2 \tau(k, t)$$

$$\text{Hence } \frac{\partial \tau}{\partial t}(k, t) = \kappa [-k^2 \tau(k, t)] = -\kappa k^2 \tau(k, t)$$

$$\text{Record the initial condition in } k\text{-space: } \tau(k, 0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} T_0(x) e^{-ikx} dx = \tau_0(k)$$

$$\text{Therefore the transformed PDE is the first-order ODE: } \frac{\partial \tau}{\partial t}(k, t) = -\kappa k^2 \tau(k, t), \quad \tau(k, 0) = \tau_0(k)$$

$$5. \text{ Solve this ODE for each fixed } k: \frac{d}{dt} \tau(k, t) = -\kappa k^2 \tau(k, t), \quad \tau(k, 0) = \tau_0(k) \implies \tau(k, t) = \tau_0(k) \exp(-\kappa k^2 t)$$

$$\tau(k, t) = \tau_0(k) e^{-\kappa k^2 t}, \quad -\infty < k < \infty, \quad t \geq 0$$

$$\text{For general initial data } T_0(x), \tau(k, t) = \tau_0(k) e^{-\kappa k^2 t}, \quad \tau_0(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} T_0(x) e^{-ikx} dx,$$

$$\longrightarrow T(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tau(k, t) e^{ikx} dk$$

$$T(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} T_0(y) \exp(-\kappa k^2 t) \exp[ik(x-y)] dy dk$$

$$= \int_{-\infty}^{\infty} T_0(y) \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} \exp(-\kappa k^2 t) \exp[ik(x-y)] dk \right] dy \quad (\text{swap integrals})$$

$$= \int_{-\infty}^{\infty} T_0(y) \underbrace{\left[\frac{1}{2\pi} \int_{-\infty}^{\infty} \exp(-\kappa k^2 t + ik(x-y)) dk \right]}_{I(x-y, t)} dy$$

$$\text{Next, set } z = x - y. \longrightarrow I(z, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp(-\kappa k^2 t + ikz) dk$$

Complete the square in the exponent:

$$-\kappa k^2 t + ikz = -\kappa t \left(k^2 - \frac{ikz}{\kappa t} \right) = -\kappa t \left[\left(k - \frac{iz}{2\kappa t} \right)^2 - \left(\frac{iz}{2\kappa t} \right)^2 \right] = -\kappa t \left(k - \frac{iz}{2\kappa t} \right)^2 - \frac{z^2}{4\kappa t}$$

$$\text{Therefore, } I(z, t) = \frac{1}{2\pi} \exp\left(-\frac{z^2}{4\kappa t}\right) \int_{-\infty}^{\infty} \exp\left[-\kappa t \left(k - \frac{iz}{2\kappa t} \right)^2\right] dk$$

$$\text{When } k' = k - \frac{iz}{2\kappa t}, \int_{-\infty}^{\infty} \exp\left[-\kappa t \left(k - \frac{iz}{2\kappa t} \right)^2\right] dk = \int_{-\infty}^{\infty} \exp(-\kappa t (k')^2) dk' = \sqrt{\frac{\pi}{\kappa t}}$$

$$\text{This tells, } I(z, t) = \frac{1}{2\pi} \exp\left(-\frac{z^2}{4\kappa t}\right) \sqrt{\frac{\pi}{\kappa t}} = \frac{1}{2\pi} \sqrt{\frac{\pi}{\kappa t}} \exp\left(-\frac{z^2}{4\kappa t}\right) = \frac{1}{\sqrt{4\pi\kappa t}} \exp\left(-\frac{z^2}{4\kappa t}\right).$$

$$\text{Since } z = x - y, \text{ this shows the inner integral equals } \frac{1}{\sqrt{4\pi\kappa t}} \exp\left(-\frac{(x-y)^2}{4\kappa t}\right).$$

$$\text{Substituting back into the outer integral } \implies T(x, t) = \int_{-\infty}^{\infty} T_0(y) \frac{1}{\sqrt{4\pi\kappa t}} \exp\left(-\frac{(x-y)^2}{4\kappa t}\right) dy$$

$$\text{Final Answer: } T(x, t) = \int_{-\infty}^{\infty} T_0(y) \frac{1}{\sqrt{4\pi\kappa t}} \exp\left(-\frac{(x-y)^2}{4\kappa t}\right) dy$$

(b)

1. Compute the initial transform:

$$\tau_0(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} T_0(x) e^{-ikx} dx = \frac{1}{\sqrt{2\pi}} \int_{-1}^1 100 e^{-ikx} dx \quad \text{Let } I(k) = \int_{-1}^1 e^{-ikx} dx$$

2. Evaluate $I(k)$ explicitly:

$$I(k) = \left[\frac{e^{-ikx}}{-ik} \right]_{x=-1}^{x=1} = \frac{1}{-ik} (e^{-ik} - e^{ik}) = \frac{1}{-ik} (-2i \sin(k)) = \frac{2 \sin(k)}{k}$$

$$\longrightarrow \tau_0(k) = \frac{100}{\sqrt{2\pi}} I(k) = \frac{100}{\sqrt{2\pi}} \frac{2 \sin(k)}{k} = \frac{200}{\sqrt{2\pi}} \frac{\sin(k)}{k}$$

3. Get $\tau(k, t)$ using the result from part (a):

$$\tau(k, t) = \tau_0(k) e^{-\kappa k^2 t} = \frac{200}{\sqrt{2\pi}} \frac{\sin(k)}{k} \exp(-10^3 k^2 t)$$

4. Invert the Fourier transform to recover $T(x, t)$:

$$T(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tau(k, t) e^{ikx} dk = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[\frac{200}{\sqrt{2\pi}} \frac{\sin(k)}{k} e^{-10^3 k^2 t} \right] e^{ikx} dk = \frac{200}{2\pi} \int_{-\infty}^{\infty} \frac{\sin(k)}{k} e^{-10^3 k^2 t} e^{ikx} dk$$

5. Convolution form using the heat-kernel. Because $T_0(y) = 100$ for $y \in [-1, 1]$ and 0 otherwise.

$$\longrightarrow T(x, t) = 100 \int_{-1}^1 \frac{1}{\sqrt{4\pi(10^3)t}} \exp\left(-\frac{(x-y)^2}{4(10^3)t}\right) dy$$

When $y \in [-1, 1] \longrightarrow T_0(y) = 100$ and $\kappa = 10^3$,

$$G(z, t) = \frac{1}{\sqrt{4\pi\kappa t}} \exp\left(-\frac{z^2}{4\kappa t}\right) \implies T(x, t) = 100 \int_{-1}^1 G(x-y, t) dy$$

$$\text{Split the integral at } y = 0: \quad T(x, t) = 100 \frac{1}{\sqrt{4\pi\kappa t}} \left[\int_{-1}^0 \exp\left(-\frac{(x-y)^2}{4\kappa t}\right) dy + \int_0^1 \exp\left(-\frac{(x-y)^2}{4\kappa t}\right) dy \right]$$

$$\text{Then we can say, } I_1 = \int_{-1}^0 \exp\left(-\frac{(x-y)^2}{4\kappa t}\right) dy \quad \text{and} \quad I_2 = \int_0^1 \exp\left(-\frac{(x-y)^2}{4\kappa t}\right) dy$$

$$\text{which leads to } \implies T(x, t) = 100 \frac{1}{\sqrt{4\pi\kappa t}} (I_1 + I_2)$$

1) Evaluation of I_1

Use the change of variable $u = \frac{x-y}{2\sqrt{\kappa t}}$, $dy = -2\sqrt{\kappa t} du$.

$$\text{When } y = -1 \longrightarrow u_1 = \frac{x-(-1)}{2\sqrt{\kappa t}} = \frac{x+1}{2\sqrt{\kappa t}} \quad \text{and when } y = 0 \longrightarrow u_0 = \frac{x-0}{2\sqrt{\kappa t}} = \frac{x}{2\sqrt{\kappa t}}$$

$$I_1 = \int_{y=-1}^0 \exp\left(-\frac{(x-y)^2}{4\kappa t}\right) dy = \int_{u=u_1}^{u=u_0} \exp(-u^2) (-2\sqrt{\kappa t}) du = 2\sqrt{\kappa t} \int_{u_0}^{u_1} e^{-u^2} du$$

$$= 2\sqrt{\kappa t} \cdot \frac{\sqrt{\pi}}{2} \left[\text{erf}(u_1) - \text{erf}(u_0) \right] = \sqrt{\pi\kappa t} \left[\text{erf}\left(\frac{x+1}{2\sqrt{\kappa t}}\right) - \text{erf}\left(\frac{x}{2\sqrt{\kappa t}}\right) \right]$$

2) Evaluation of I_2

Use the change of variable $u = \frac{x-y}{2\sqrt{\kappa t}}$, $dy = -2\sqrt{\kappa t} du$

As y runs from 0 to 1, when $y = 0 \longrightarrow u_0 = \frac{x}{2\sqrt{\kappa t}}$ and when $y = 1 \longrightarrow u_2 = \frac{x-1}{2\sqrt{\kappa t}}$.

$$I_2 = \int_{y=0}^1 \exp\left(-\frac{(x-y)^2}{4\kappa t}\right) dy = \int_{u=u_0}^{u=u_2} \exp(-u^2) (-2\sqrt{\kappa t}) du = 2\sqrt{\kappa t} \int_{u_2}^{u_0} e^{-u^2} du$$

$$= 2\sqrt{\kappa t} \cdot \frac{\sqrt{\pi}}{2} \left[\text{erf}(u_0) - \text{erf}(u_2) \right] = \sqrt{\pi\kappa t} \left[\text{erf}\left(\frac{x}{2\sqrt{\kappa t}}\right) - \text{erf}\left(\frac{x-1}{2\sqrt{\kappa t}}\right) \right]$$

Combine $I_1 + I_2$:

$$I_1 + I_2 = \sqrt{\pi\kappa t} \left[\text{erf}\left(\frac{x+1}{2\sqrt{\kappa t}}\right) - \text{erf}\left(\frac{x}{2\sqrt{\kappa t}}\right) \right] + \sqrt{\pi\kappa t} \left[\text{erf}\left(\frac{x}{2\sqrt{\kappa t}}\right) - \text{erf}\left(\frac{x-1}{2\sqrt{\kappa t}}\right) \right]$$

$$I_1 + I_2 = \sqrt{\pi\kappa t} \left[\text{erf}\left(\frac{x+1}{2\sqrt{\kappa t}}\right) - \text{erf}\left(\frac{x-1}{2\sqrt{\kappa t}}\right) \right]$$

$$\text{Therefore, } T(x, t) = 100 \frac{1}{\sqrt{4\pi\kappa t}} (I_1 + I_2) = 100 \frac{1}{\sqrt{4\pi\kappa t}} \sqrt{\pi\kappa t} \left[\text{erf}\left(\frac{x+1}{2\sqrt{\kappa t}}\right) - \text{erf}\left(\frac{x-1}{2\sqrt{\kappa t}}\right) \right]$$

Since $\frac{\sqrt{\pi \kappa t}}{\sqrt{4\pi \kappa t}} = \frac{1}{2} \implies T(x, t) = 50 \left[\operatorname{erf}\left(\frac{x+1}{2\sqrt{10^3 t}}\right) - \operatorname{erf}\left(\frac{x-1}{2\sqrt{10^3 t}}\right) \right]$

Final Answer: $T(x, t) = 50^\circ\text{C} \left[\operatorname{erf}\left(\frac{x+1}{2\sqrt{10^3 t}}\right) - \operatorname{erf}\left(\frac{x-1}{2\sqrt{10^3 t}}\right) \right] \quad |x| \leq 1m, t > 0$

Problem 5

